PERFORM QUALITY CONTROL

Introduction

Quality control (QC) for laboratory equipment ensures that diagnostic tests and procedures are accurate, reliable, and consistent. QC helps prevent errors, maintain equipment performance, and ensure patient safety. If quality controls are not done or are inaccurate, this can lead to invalid or inaccurate test results. Inaccurate test results may require extra testing and delay in diagnosis and treatment, which can cause both mental and physical distress to the patient.

quality control (QC). The process that is used to ensure the accuracy and reliability of results.

Quality Control

Laboratory Equipment

Quality control ensures accuracy and reliability of results, which is essential in the laboratory, especially in the equipment used to perform the testing. Organizations such as the CLSI and the CDC set guidelines for quality control procedures. The guidelines ensure that both equipment and procedures align with standards, protect patient outcomes, and maintain the accuracy of laboratory results. QC done on laboratory equipment includes regular calibration, maintenance, and performance verification. Each facility will have guidelines on how frequently quality controls must be done for the equipment and testing supplies. Each time quality controls are run on the equipment and testing materials, the phlebotomist performing the quality controls will need to document the outcome and any action taken to fix or correct inaccurate results on the quality control. The QC logs can indicate to agencies such as TJC and the CLSI that QCs are being done and can indicate when routine maintenance and calibration are occurring. These logs also allow other employees in the laboratory to check that QCs have been done. The Clinical Laboratory Improvement Advisory Committee (CLIAC) was formed to assist in administering regulations and advise if revisions are needed in the current standards.

2.2 CHALLENGE

1. How does a phlebotomy technician's attention to detail, particularly in identifying and addressing minor inconsistencies during quality controls, impact patient safety and the integrity of laboratory results?

2.2 CHALLENGE KEY

1. A phlebotomy technician's meticulous attention to detail is crucial in safeguarding patient safety and ensuring the reliability of laboratory results. By noticing and rectifying even minor inconsistencies during quality controls, the technician prevents the possibility of erroneous test outcomes that could lead to misdiagnosis or inappropriate treatment plans.

Quality Control Versus Quality Assurance

Quality control and quality assurance (QA), which are often used interchangeably, are not the same. QA is a process designed to prevent future problems by analyzing past and present performance. QC is a factor in QA and is used to ensure the accuracy of results before they are finalized and released to the provider and patient.

Professionalism Tip

Attention to detail for a phlebotomy technician involves ensuring that every facet of their work, from blood collection to recordkeeping, is precise and free from mistakes. For instance, a phlebotomy technician might observe a minor inconsistency when running quality controls, prompting them to confirm the details and correct the error to avert any potential issues with inaccurate results.